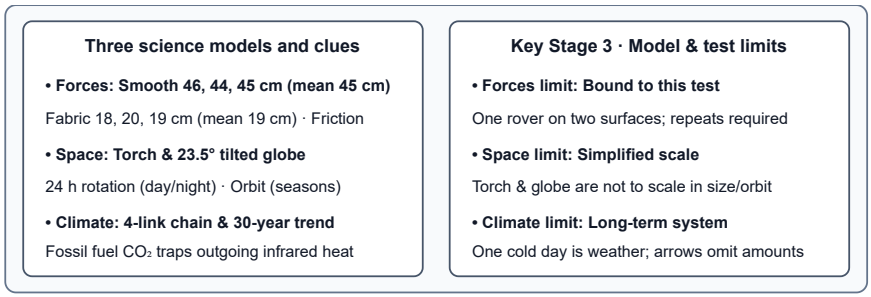
# Reference sheets

## Reference A · Three science models, clues, reasons, and limits

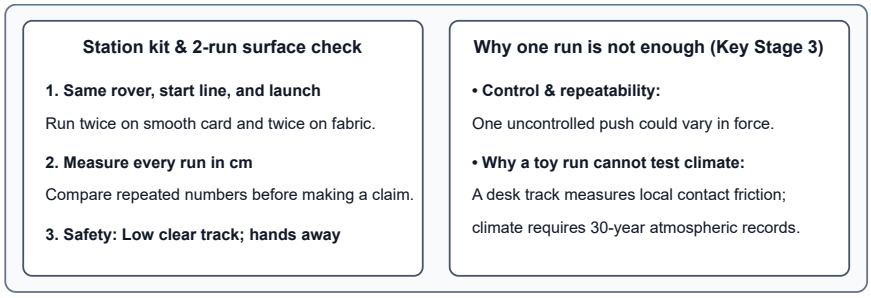



Three science models diagram. It shows Forces surface runs, Space torch and globe, and Climate warming chain with limits.

Reference only. Do not write on this sheet. Read the question first, then pick the matching model row.

| Science question. | Model and evidence clue. | Key Stage 3 science reason. | Model or test limit. |
|---|---|---|---|
| 1. Forces: Which surface lets the rover travel farther? | Rover track: Smooth 46, 44, 45 cm (mean 45 cm). Fabric 18, 20, 19 cm (mean 19 cm). | Contact friction opposes motion between rover wheels and the surface. | Only one rover and two surfaces were tested ('in this test'). |
| 2. Space: Why is a location in darkness or winter? | Fixed torch and turning 23.5°-tilted globe: marked spot turns into shadow. | Earth's 24-hour rotation causes day and night. Its 23.5° tilt during orbit causes seasons. | The classroom torch and globe are not to scale ('in this model'). |
| 3. Climate: Why does extra $CO_2$ warm Earth over decades? | Four-link warming chain and 30-year climate trend. | Fossil fuel $CO_2$ traps outgoing infrared heat. Mitigation cuts gas (Link 1); adaptation cuts harm (Link 4). | One cold day is weather, not climate. Arrows omit exact amounts. |

## Reference B · Desk station check and safe two-run rover method




Desk station check diagram. It shows the two-run rover method, safety rules, and test limits.

| Station step. | What we do at the desk. | Key Stage 3 reason and safety rule. |
|---|---|---|
| 1. Match kit to question. | Use rover + surfaces for motion, torch + globe for day/night, and warming cards for climate. | A model can only test a question that matches its mechanism. |
| 2. Fair two-run rover check. | Keep the same rover, start line, and push. Measure two runs per surface in cm. | One uncontrolled run cannot prove a cause. Repeats check if the pattern holds. |
| 3. Safety on the desk. | Use a low clear track. Keep hands and faces away. Keep torch light out of eyes. | Prevents the moving toy falling or hitting anyone. |

**Made by Matt**

# Tier pack · Independent

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

## Arrival · 4 questions

**1.** Use the arrival diagram. Match each question (surface travel, day and night, warming) to its model in Boxes 1 to 3.

*Help. Pair each question with the model that shows its clue.*

**2.** Look at '46 cm across three trials'. Why is it stronger evidence than saying '46' or 'I like smooth card'?

*Help. Check for a measurement unit and repeated runs.*

**3.** Look at Box 2 of the arrival diagram. How does the torch and tilted globe model explain both day/night and seasons?

*Help. Separate 24-hour spin from the 23.5° axial tilt.*

**4.** Look at Box 3 of the arrival diagram. Why does extra carbon dioxide warm Earth over decades?

*Help. Link fossil fuel $CO_2$ to outgoing infrared heat.*

## You do tasks

**1. Use Reference A. Choose two science questions (from Forces, Space, and Climate). For each question, state the matching model, the evidence clue, and the science reason.** **Staff I ☐ S ☐ C ☐**

*Help. Include contact friction, 24-hour rotation or 23.5° tilt, or $CO_2$ trapping infrared heat.*

**2. Use Reference A. Why must 'Earth rotates in 24 hours' be rejected as evidence for the rover question?** **Staff I ☐ S ☐ C ☐**

*Help. Compare the mechanism in the question with the mechanism in the clue.*

**3. Use Reference A and Reference B. State one clear limit for each of the two models you chose. Why do we add 'in this test' or 'in this model'?** **Staff I ☐ S ☐ C ☐**

*Help. Check the Model limit column in Reference A.*

**Optional participation-loop prompt.** Share your model match and clue with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

## Exit ticket

**1.** Match each question (rover travel, day and night, global warming) to its model and clue.

**2.** Look at the distance table (46, 44, 45 cm vs 18, 20, 19 cm). Why is it relevant evidence for the surface question?

**3.** State one limit of the rover surface test. Or state one limit of the globe model.

**Pupil voice**

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

**Feedback and response · Independent**

## Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

| Code | Meaning |
|---|---|
| VF | Verbal feedback |
| WS | Worked with support |
| I | Independent |
| NS + … | Next step |
| E | Evidence captured |
| R | Pupil responded to feedback |
| // | Self-edit point |
| ? | Read this back to me |

## My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

**Made by Matt**